

Selective Betting with Shrinkage

Final metrics, 27 September 2026. Every figure is out-of-sample; brackets give the 5th–95th percentile across random seeds.

Data. 7,466 matches, 22 European divisions, season 2025/26.

Features. 66 per match, none derived from betting prices.

Market. Average closing odds, margin removed with Shin's method.

Validation. Walk-forward over 10 chronological blocks: 8 test folds.

Repetition. 100 random seeds per configuration (800 fits).

Bets. Top 10% by edge; settled at the best closing price.

1 Probability quality: log-loss

Log-loss measures how much probability a model gave to the result that actually happened; lower is better, and it is the metric every model is ranked on. The betting band is the 10% of matches where the model disagrees most with the market, which is where bets are placed.

Model	All matches	Betting band
Uniform guess (1/3 each)	1.099	—
Base rate (always home win)	1.075	—
Market (closing odds)	0.995	0.907–0.923
Stage 1, MLP	1.022	1.011
Stage 1, CatBoost	1.018	0.996
Stage 1, MLP ensemble of 5	1.019	0.992
Corrected (shrunk toward the market)	0.995	= market

The model clearly learns football (1.02 against 1.075 for the base rate) but never reaches the market. Its gap to the market is largest in the betting band, exactly where it would bet. The market's band score varies slightly because each model's band contains different matches.

2 Accuracy

Accuracy is the share of matches in which the outcome given the highest probability actually happened. It is reported for completeness but not used for ranking, because betting depends on how well calibrated the probabilities are, not on picking the most likely result.

Model	Accuracy
Market (closing odds)	51.5%
Stage 2 (market-aware comparison model)	50.9%
Stage 1 (market-blind model)	48.9%
Base rate (always home win)	43.5%

Single run (seed 42, MLP, 5,808 test matches). Neither the model nor the market almost ever predicts a draw, since a draw is rarely the single most likely result.

3 Shrinkage weight w

The corrected probability is $p_{\text{corr}} \propto p_{\text{market}}^{1-w} \cdot p_{\text{model}}^w$, with w fitted to minimise log-loss: $w = 0$ means "trust only the market", $w = 1$ "trust only the model". w_{band} is the same weight fitted only on the betting band, to check whether an edge hides where bets are placed.

Configuration	w median [95th pct]	Fits with $w < 0.05$	w_{band} median [95th pct]
MLP	0.000 [0.046]	96%	0.000 [0.083]
CatBoost	0.000 [0.000]	100%	0.000 [0.040]
MLP ensemble of 5	0.000 [0.010]	100%	0.000 [0.041]

In almost all 2,400 fits the data prefers the market to the model, including where bets are placed. This is the central finding: the model's opinion adds nothing the closing price does not already contain.

4 Betting results

The operating point bets the 10% of candidate home and away bets with the largest edge (model probability minus market probability), using a threshold learned on validation data. Because the corrected model collapses onto the market, bets are placed on the uncorrected model's opinion.

Configuration	ROI median	Best seed	Seeds > 0	P(ROI ≤ 0)	Beats random
MLP	−12.2% [−16.9, −6.1]	−3.7%	0%	0.97	43%
CatBoost	−12.9% [−16.8, −9.0]	−7.3%	0%	n/a	30%
MLP ensemble of 5	−10.4% [−14.8, −6.7]	−5.3%	0%	0.94	55%

ROI median

Profit per unit staked over the season, the middle value across 100 seeds. Every configuration loses money.

Best seed

The single best season result out of 100 runs. Even the luckiest run loses.

Seeds > 0

Share of the 100 seeds with a profitable season. None were profitable, so the losses are systematic rather than bad luck.

P(ROI ≤ 0)

Probability of a loss from a bootstrap that resamples whole weeks, since matches on the same weekend are correlated. Values near 1 mean the loss is statistically clear.

Beats random

Share of random bet selections with the same odds profile that the model beats. 50% is chance, so the model's bet selection is no better than random.

5 Effect of the feature set

The same MLP was run with 26 features (form and shots only) and with all 66 (adding Elo ratings, last season's table, an expected-goals proxy and league context). This isolates the value of information from the value of the learner.

Metric	26 features	66 features
Stage-1 log-loss	1.042	1.022
ROI median	−20.7%	−12.2%
Beats random	15%	43%

More information halved the loss, while changing the learner (MLP, CatBoost, ensemble; section 4) changed it by only about 2 points. The limit is information, not model capacity.

6 Log-loss as a function of w (seven seasons)

Log-loss of the blended probabilities for fixed values of w on the most recent fold of the seven-season dataset. It shows the whole curve the fitted weight is chosen from, rather than only its minimum.

w	0.0	0.1	0.3	0.5	1.0
All matches	0.998	0.999	1.001	1.005	1.022
Betting band	0.894	0.896	0.904	0.922	1.009

Every step away from $w = 0$ makes the forecasts worse, on all matches and on the betting band. Scored on the fitting window, not a held-out season.

Bottom line

The market-blind model is a genuine predictor, clearly better than the base rate, but in every configuration it remains worse than the closing odds, and most of all where it would bet.

The fitted shrinkage weight therefore stays at zero, and betting on the model's disagreements loses 10–13% in every one of 300 runs. This indicates that public match features carry no information beyond the closing odds.